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FSK 176

Semester Test 1 - Semestertoets 1

Department of Physics, University of Pretoria
Departement Fisika, Universiteit van Pretoria

Date / Datum: 27/08/2016

Time / Tyd: 2.5 hours / ure

Total / Totaal: 65 max

Examiner / Eksaminator: Dr J M Nel

Moderator / Moderator: R Q Odendaal

Test register: Tear carefully on the line above and hand to the invigilator.

Toets register: Skeur versigtig op die lyn hierbo en handig dit in by die toesighouer.

FSK 176: Semester Test 1 - Semestertoets 1

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Please read the following carefully. *Lees die volgende noukeurig*

1. Answer all the questions in the space provided on the question paper. *Antwoord alle vrae in die spasie gelaat op die vraestel.*
2. No pencil work will be marked. Do not use "tippex". *Geen potlood-werk sal gemerk word nie. Moet nie uitwis-vloeistof ("tippex") gebruik nie.*
3. If you need more space, please use the back of the pages, and clearly indicate where the question continues. *As u meer spasie benodig gebruik die agterkant van die bladsye. Dui duidelik aan waar die vraag aangaan.*
4. Where applicable you need to draw the necessary diagrams (e.g. free-body diagrams). *Teken diagramme waar nodig (bv. vryliggaams-diagramme).*
5. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. **Only substitute numerical values in the final step of calculations! A correct answer does not imply that you will receive full marks.** *U antwoorde moet sistematies en logies neergeskryf word. U moet bewys dat u die nodige fisika verstaan en kan toepas. Numeriese waardes mag slegs in die laaste stap van 'n berekening vervang word! 'n Korrekte antwoord beteken nie dat u noodwendig vol punte sal kry nie.*
6. Units must be used when numbers are substituted else marks will be subtracted. *Eenhede moet gebruik word wanneer numeriese waardes vervang word, andersins sal punte afgetrek word.*
7. The use of the cheat sheet is only allowed subject to the rules stated on it. *Die formuleblad mag slegs gebruik word onderhewig aan die reëls daarop gedruk.*
8. All test/exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner. *Alle toets/eksamen regulasies van die Universiteit van Pretoria geld en enige oortreding daarvan sal in 'n ernstige lig beskou word.*

Constants and formulas / Konstantes en formules

$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$; $G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$; $e = 1.602 \times 10^{-19} \text{ C}$; $k = 1.381 \times 10^{-23} \text{ J/K}$; $m_e = 9.109 \times 10^{-31} \text{ kg}$; $m_p = 1.673 \times 10^{-27} \text{ kg}$; $R = 8.314 \text{ J}/(\text{mol}\cdot\text{K})$; $N_A = 6.02 \times 10^{23} / \text{mol}$; $1 \text{ atm} = 101.3 \text{ kPa}$; $1 \text{ u} = 1.6605677 \times 10^{-27} \text{ kg}$; $c = 3 \times 10^8 \text{ m/s}$; $V_{\text{sphere/sfeer}} = \frac{4}{3}\pi r^3$; $F_{\text{gravity}} = \frac{Gm_1m_2}{r^2}$; $1 \text{ litre} = 1000 \text{ cm}^3$; $M_{\text{Earth}} = M_{\text{Aarde}} = 5.97 \times 10^{24} \text{ kg}$; $r_{\text{Earth}} = r_{\text{Aarde}} = 6.37 \times 10^6 \text{ m}$; $c_{\text{ice}} = 2090 \text{ J/kg}\cdot^\circ\text{C}$; $c_{\text{water}} = 4186 \text{ J/kg}\cdot^\circ\text{C}$; $c_{\text{steam}} = 2010 \text{ J/kg}\cdot^\circ\text{C}$; $L_{\text{water},f} = 3.33 \times 10^5 \text{ J/kg}$; $L_{\text{water},v} = 2.26 \times 10^6 \text{ J/kg}$; $\rho_{\text{water},0^\circ\text{C}} = 0.999 \text{ g/cm}^3$, $\rho_{\text{water},4^\circ\text{C}} = 1.000 \text{ g/cm}^3$

1 Question 1 / Vraag 1

[5]

A partly full paint can has 3.00 litres of paint left in it.
'n Verfblik is gedeeltelik vol en het 3.00 liter verf in dit.

- (a) What is the volume of the paint in cubic millimetres? Show all conversion factors used in your calculation.

Wat is die volume van die verf in kubieke millimeter? Toon alle omskakelingsfaktore wat jy gebruik in jou berekening.

(2)

$$\begin{aligned}
 1 \text{ litre} &= 1000 \text{ cm}^3 \\
 3.00 \text{ litre} &= 3.00 \text{ litre} \left(\frac{1000 \text{ cm}^3}{1 \text{ litre}} \right) \left(\frac{10 \text{ mm}}{1 \text{ cm}} \right)^3 \\
 &= 3.00 \times 10^6 \text{ mm}^3
 \end{aligned}$$

- (b) If all the remaining paint is used to coat a wall evenly (wall area is 15 m^2), how thick is the layer of wet paint? Give your answer in micrometres.

Indien al die verf gebruik word om 'n muur eweredig te verf (die muur area is 15 m^2), hoe dik is die laag nat verf? Gee jou antwoord in mikrometer. (3)

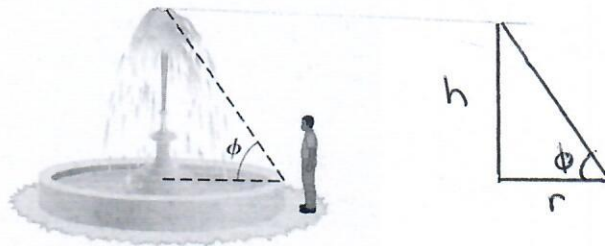
$$\begin{aligned} \text{Volume} &= \text{Area} \times \text{thickness} \\ \text{thickness} &= \frac{\text{Volume}}{\text{Area}} \\ &= \frac{3.00 \times 10^6 \text{ mm}^3}{15 \text{ m}^2 \left(\frac{1000 \text{ mm}}{1 \text{ m}} \right)^2} \\ &= 200 \times 10^{-3} \text{ mm} \times \left(\frac{1000 \mu\text{m}}{1 \text{ mm}} \right) \\ &= 200 \mu\text{m} \end{aligned}$$

2 Question 2 / Vraag 2

[4]

A high fountain of water is located at the centre of a circular pool, as shown in the figure. A student walks around the pool and measures its circumference, C . Next, the student stands at the edge of the pool and uses a protractor to measure the angle of elevation, ϕ , of the top of the fountain. Find an expression for the height of the fountain in terms of C and ϕ .

'n Hoë water-fontein staan in die middle van 'n ronde dam. Die student stap om die dam en meet die omtrek, C . Daarna staan die student by die rant van die dam en meet die hoogte-hoek, ϕ van die bo-punt van die fontein met 'n gradeboog. Vind 'n vergelyking vir die hoogte van die fontein in terme van C en ϕ .



$$\begin{aligned} \text{Circumference: } C &= 2\pi r \quad \text{--- ①} \\ \text{Radius of pool: } &r \\ \text{From triangle above: } \tan \phi &= \frac{h}{r} \\ \therefore h &= r \tan \phi \\ \text{From ① } r &= \frac{C}{2\pi} \\ \therefore h &= \frac{C}{2\pi} (\tan \phi) \end{aligned}$$

3

Question 3 / Vraag 3

[6]

When the acceleration of a particle moving in the x -direction is constant, the average and instantaneous acceleration are equal: $a_x = a_{x,avg} = \frac{v_{xf} - v_{xi}}{t - 0}$, where a_x is the acceleration, v_{xf} is the velocity at time t and v_{xi} is the initial velocity at $t = 0$. We can also write the average velocity as $v_{x,avg} = \frac{x_f - x_i}{t - 0}$, where x_f is the position at time t and x_i is the initial ^{Position} velocity at $t = 0$. Using these two definitions and **integration** show that when a particle experiences constant acceleration, its displacement is given by $x_f - x_i = v_{xi}t + \frac{1}{2}a_x t^2$. Show all your steps and state your reasoning clearly.

Wanneer die versnelling konstant is, is die gemiddelde en oombliklike versnelling in die x -rigting dieselfde: $a_x = a_{x,gem} = \frac{v_{xf} - v_{xi}}{t - 0}$, waar a_x die versnelling is, v_{xf} die snelheid by tyd t en v_{xi} die aanvanklike snelheid by $t = 0$ is. Ons kan ook die gemiddelde snelheid as $v_{x,gem} = \frac{x_f - x_i}{t - 0}$ skryf, waar x_f die posisie by tyd t en x_i die aanvanklike posisie by $t = 0$ is. Deur gebruik te maak van hierdie twee definisies en **integrasië** bewys dat die verplasing van 'n deeltjie gegee word deur $x_f - x_i = v_{xi}t + \frac{1}{2}a_x t^2$ indien die deeltjie konstante versnelling ondergaan. Wys al jou stappe en verduidelik jou aannames.

$$a_x = \frac{v_{xf} - v_{xi}}{t - 0} = \frac{\Delta v_x}{\Delta t}$$

$$\text{As } \Delta t \rightarrow 0 \quad a_x = \frac{dv_x}{dt}$$

$$\therefore a_x dt = dv_x$$

$$\text{Integrating} \int_0^t a_x dt = \int_{v_{xi}}^{v_{xf}} dv_x$$

If you don't have limits you must have integration constants

$$a_x(t - 0) = v_{xf} - v_{xi} \quad \text{--- (1)}$$

$$\text{Similarly } v_x = \frac{\Delta x}{\Delta t} \text{ and as } \Delta t \rightarrow 0 \quad v_x = \frac{dx}{dt}$$

$$\therefore dx = v_x dt$$

$$\text{Integrating} \int_{x_i}^{x_f} dx = \int_0^t v_x dt \quad \text{--- (2)}$$

also here.

$$\text{But } v_x = v_{xf} \text{ then } v_x = v_{xi} + at \quad \text{from (1)}$$

$$\begin{aligned} \text{Substitute in (2)} \quad x_f - x_i &= \int_0^t (v_{xi} + at) dt \\ &= \left[v_{xi}t + \frac{1}{2}at^2 \right]_0^t \\ &= v_{xi}t + \frac{1}{2}a_x t^2 \end{aligned}$$

4

Question 4 / Vraag 4

[12]

A car travelling at 15 m/s has to slow to a speed of 5 m/s to get over a speed bump. It starts to slow down at a constant rate 5 s before reaching the bump, takes 1 s to traverse the bump and 5 s to accelerate back up to 15 m/s.

'n Kar wat teen 15 m/s beweeg moet sy spoed verminder tot 5 m/s om oor 'n spoedwal te ry. Dit begin teen 'n konstante tempo stadiger beweeg, 5 s voordat dit die wal bereik; dit neem 1 s om die wal oor te gaan, en 5 s om te versnel tot 15 m/s.

- (a) What is the car's initial speed in km/h?

Wat is die kar se aanvanklike spoed in km/h?

(3)

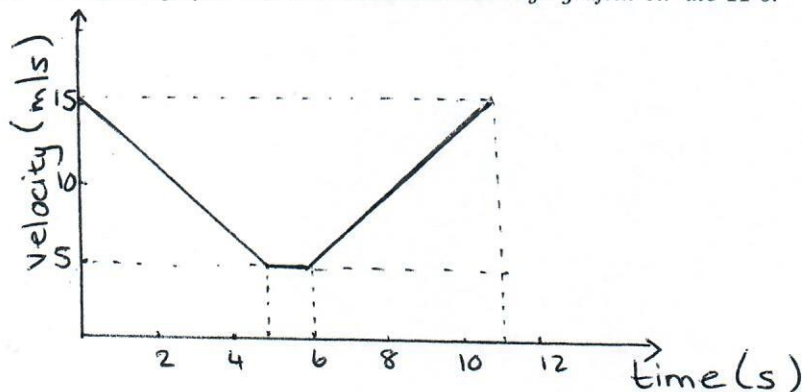
$$15 \text{ m/s} = \frac{15 \text{ m}}{\text{s}} \left(\frac{1 \text{ km}}{1000 \text{ m}} \right) \left(\frac{60 \text{ s}}{1 \text{ min}} \right) \left(\frac{60 \text{ min}}{1 \text{ hr}} \right)$$

$$= 54 \text{ km/h}$$

- (b) Sketch a neat graph of the velocity vs time for all 11 s.

Skets 'n netjiese grafiek van die snelheid teenoor tyd grafiek vir die 11 s.

(3)



axes
shape

- (c) How far does car travel during the 11 s?

Hoe ver beweeg die kar in die 11 s?

(3)

Distance travelled = Area under the curve

$$\Delta x = \left[\frac{1}{2} (10 \text{ m/s}) (5 \text{ s}) + (5 \text{ m/s}) (5 \text{ s}) \right] \times 2 +$$

$$(5 \text{ m/s}) (1 \text{ s})$$

$$= 100 \text{ m} + 5 \text{ m}$$

$$= 105 \text{ m}$$

Can also use formulas

Need to calculate "a" first

- (d) By how many seconds has the bump delayed the progress of the car? (in other words, how much longer does it take to travel the distance in (c) compared to when there is no speed bump present?)
 Met hoeveel sekondes het die wal die vordering van die kar vertraag? (Met ander woorde, hoeveel langer vat dit om die afstand in (c) af te lê in vergelyking met wanneer daar nie 'n speedwal is nie. (3)

travelling at 15 m/s:

$$\Delta x = v_{ox} t$$

$$t = \frac{\Delta x}{v_{ox}} = \frac{105 \text{ m}}{15 \text{ m/s}} = 7 \text{ s.}$$

Progress has been delayed by $11 \text{ s} - 7 \text{ s} = 4 \text{ s.}$

5 Question 5 / Vraag 5

[8]

A particle's velocity is given by $v_z(t) = -(15.00 \text{ m/s}^2)t$ for $t \geq 0$.
 'n Deeltjie se snelheid word gegee deur $v_z(t) = -(15.00 \text{ m/s}^2)t$ vir $t \geq 0$.

- (a) Find an expression for the particle's position as a function of time.
 Vind 'n vergelyking vir die deeltjie se posisie as 'n funksie van tyd. (2)

$$z = \int v dt = \int (-15.00 \text{ m/s}^2)t dt$$

$$= -7.50 \text{ m/s}^2 t^2 + c$$

OR $z = \int_0^t v dt = \int_0^t -(15.00 \text{ m/s}^2)t dt$

$$= -7.50 \text{ m/s}^2 t^2 - 0$$

- (b) Is the particle speeding up, slowing down, or maintaining a constant speed? Give a reason for your answer.

Is die deeltjie besig om vinniger, stadiger of teen 'n konstante spoed te beweeg? Gee 'n rede vir jou antwoord. (2)

Speeding up since v_z is a function of time and as t increases v_z also increases in magnitude.

- (c) What are the particle's position, velocity and acceleration at $t = 6.50$ minutes?

Wat is die deeltjie se posisie, snelheid en versnelling by $t = 6.50$ minute?

(4)

$$t = 6.50 \text{ minutes} = 6.50^{\text{min}} \times (60 \text{ s/min})$$

$$t = 390 \text{ s}$$

$$z = -7.50 \text{ m/s}^2 (390 \text{ s})^2 + C = -1.14 \times 10^6 \text{ m} + C$$

$$v_z = -15.00 \text{ m/s}^2 (390) = -5.85 \times 10^3 \text{ m/s}$$

$$a_z = dv_z/dt = -15.00 \text{ m/s}^2$$

6 Question 6 / Vraag 6

[8]

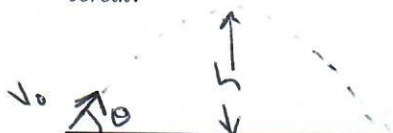
An archer shoots an arrow in a direction 30° above the horizontal with a speed of 50 m/s .

'n Boogskietster skiet 'n pyl in die rigting 30° bo die horisontale rigting teen 'n spoed van 50 m/s .

- (a) Determine the maximum height reached by the arrow. How long does it take to reach this maximum height?

Bereken die maksimum hoogte wat die pyl bereik. Hoe lank neem dit om dié maksimum hoogte te bereik?

(4)



$$v_0 = 50 \text{ m/s}$$

$$\theta = 30^\circ$$

At max height, h , $v_y = 0 \text{ m/s}$.

$$v_{iy} = 50 \text{ m/s} \sin 30^\circ = 25 \text{ m/s}$$

$$v_{ix} = 50 \text{ m/s} \cos 30^\circ = 43 \text{ m/s} \quad (\checkmark) \text{ part (b)}$$

$$\therefore v_y^2 = v_{iy}^2 + 2a\Delta y$$

$$\Delta y = (v_y^2 - v_{iy}^2) / 2a$$

$$= \frac{0 - (25 \text{ m/s})^2}{2(-9.8 \text{ m/s}^2)}$$

$$= 32 \text{ m}$$

$$v_y = v_{y0} - gt$$

$$t = \frac{v_y - v_{y0}}{-g} = 2.55 \text{ s}$$

- (b) Calculate the magnitude and direction of the velocity of the arrow after 4.0 s.
Bereken die grootte en rigting van die snelheid van die pyl na 4.0 s.

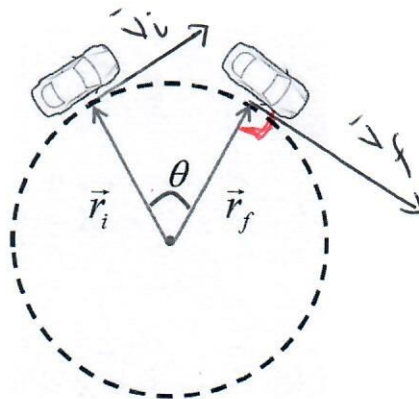
(4)

$$\begin{aligned} \text{After } 4.0 \text{ s, } v_x &= v_{ix} = 43 \text{ m/s.} \\ v_y &= v_{iy} + at = 25 \text{ m/s} + (-9.8 \text{ m/s}^2)(4.0 \text{ s}) \\ &= 25 \text{ m/s} - 39.2 \text{ m/s} \\ &= -14 \text{ m/s} \quad \checkmark \\ |v| &= \sqrt{v_y^2 + v_x^2} = \sqrt{(43 \text{ m/s})^2 + (-14 \text{ m/s})^2} \\ &= 45 \text{ m/s} \quad \checkmark \\ \theta &= \tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}\left(\frac{-14 \text{ m/s}}{43 \text{ m/s}}\right) = 18^\circ \quad \checkmark \text{ below } +x \text{ axis} \quad \checkmark \\ &\quad (341.8^\circ) \end{aligned}$$

7 Question 7 / Vraag 7

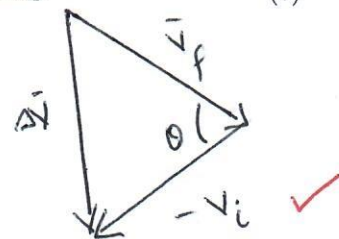
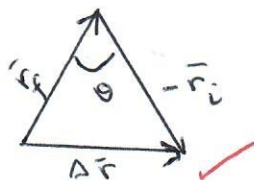
[13]

A car moves clockwise around a circular track of radius r at a constant speed.
'n Kar beweeg kloksgewys om 'n sirkelbaan met 'n radius r teen 'n konstante spoed.



- (a) Using this information and the sketch, start with the definition of average acceleration and show that $a_c = \frac{v^2}{r}$. Show all your steps and explain each step.
Deur gebruik te maak van dié inligting en die skets, begin met die definisie van gemiddelde versnelling en bewys dat $a_c = \frac{v^2}{r}$. Wys al jou stappe en verduidelik elke stap

(6)



In figures the position vectors have been drawn showing $\Delta \vec{r} = \vec{r}_f - \vec{r}_i$ and similarly with the velocity vectors showing $\Delta \vec{v} = \vec{v}_f - \vec{v}_i$

The two Δ s are similar therefore

$$\frac{|\Delta \vec{v}|}{v} = \frac{|\Delta \vec{r}|}{r} \quad \text{where } v = |\vec{v}_i| = |\vec{v}_f| \text{ and } r = |\vec{r}_i| = |\vec{r}_f|$$

$$\text{and } |\Delta \vec{v}| = \frac{v |\Delta \vec{r}|}{r}$$

By definition

$$|a_{\text{avg}}| = \frac{|\Delta \vec{v}|}{\Delta t} = \frac{v |\Delta \vec{r}|}{r \Delta t}$$

$$\text{As } \Delta t \rightarrow 0 \quad \frac{|\Delta \vec{r}|}{\Delta t} \rightarrow v$$

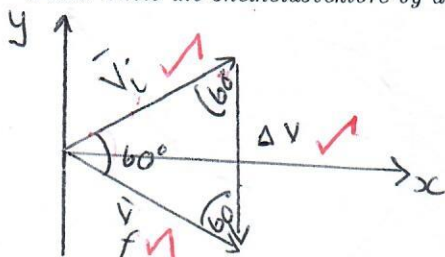
$$\therefore a_c = \frac{v^2}{r}$$

- (b) If the car is moving at a speed of 50 km/h and the radius of the circular track is 250 m, calculate the resultant **change** in the velocity when the car moves around an arc of $\theta = 60^\circ$. Hint: Sketch the velocity vector at the initial and final positions.

Indien die kar teen 'n spoed van 50 km/h beweeg en die radius van die sirkelbaan 250 m is, bereken die resultante **verandering** in die snelheid wanneer die kar om 'n sirkelboog van $\theta = 60^\circ$ beweeg.

Wenk: Skets die snelheidsvektore by die aanvangs- en finale posisies.

(4)



Considering the $\vec{v}_i$ & $\vec{v}_f$ as shown on the axes above. This is an isosceles Δ but actually turns out to be an equilateral Δ .

$\therefore |\Delta v| = 50 \text{ km/h}$ and it is directed at 60° to the $\vec{v}_i$ (original velocity)

- (c) How long does it take the car to move through the arc of $\theta = 60^\circ$.

Hoe lank neem die kar om om die sirkelboog van $\theta = 60^\circ$ te beweeg.

(3)

60° of the 360° circle is $\frac{1}{6}$ of a revolution

$$\text{Distance of arc} = \frac{1}{6} \text{ circumference} = \frac{1}{6} (2\pi)(r)$$

$$= \frac{1}{6} (2\pi)(250 \text{ m})$$

$$= 262 \text{ m}$$

$$v = \frac{\text{distance}}{\text{time}}$$

$$\text{time} = \frac{\text{distance}}{\text{speed}} = \frac{262 \text{ m}}{50 \text{ km/h} \times \left(\frac{1000 \text{ m}}{1 \text{ km}}\right) \left(\frac{1 \text{ h}}{60 \text{ min}}\right) \left(\frac{\text{min}}{60 \text{ s}}\right)}$$

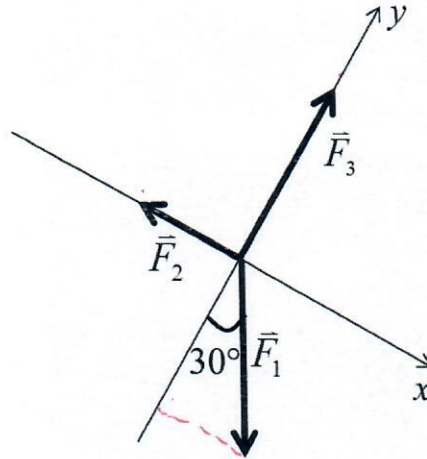
$$= 18 \text{ s}$$

8 Question 8 / Vraag 8

[4]

Three vectors are shown in the figure below, but they are not drawn to scale. The sum of the three vectors is $\vec{R} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$. If $R_y = 0$ and $|\vec{F}_2| = 0.5 |\vec{F}_3|$, find R_x in terms of $|\vec{F}_1|$.

Drie vektore word aangedui in die figuur, maar hulle is nie volgens skaal geteken nie. Die som van die drie vektore is $\vec{R} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$. Indien $R_y = 0$ en $|\vec{F}_2| = 0.5 |\vec{F}_3|$, vind R_x in terme van $|\vec{F}_1|$.



$$R_y = F_{3y} + F_{1y} + F_{2y} = F_3 - F_1 \cos 30^\circ = 0 \quad \checkmark$$

$$\therefore F_3 = F_1 \cos 30^\circ$$

$$R_x = F_{1x} + F_{2x} + F_{3x} = F_1 \sin 30^\circ - F_2 \quad \checkmark F_2 = \frac{1}{2} F_3$$

$$= F_1 \sin 30^\circ - 0.5 F_3 \quad \checkmark$$

$$R_x = F_1 \sin 30^\circ - 0.5 F_1 \cos 30^\circ \quad \checkmark$$

$$R_x = 0.067 F_1 \quad \checkmark$$

Work symbolically and systematically showing clearly what/why for each step.

9

Question 9 / Vraag 9

[6]

In a marine-rescue exercise, a small boat moves at a velocity of 13 km/h, 37° north of east. The rescue boat has a speed of 30 km/h and is positioned 42° south of east at a distance of 60 m from the small boat. In what direction must the rescue boat head to intercept the small boat?

In 'n see-reddingsoefening, beweeg 'n bootjie met 'n snelheid van 13 km/h in 'n rigting 37° noord van oos. Die reddingsboot het 'n spoed van 30 km/h en is by 'n posisie 60 m (42° suid van oos) geleë vanaf die bootjie. In watter rigting moet die reddingsboot beweeg om die bootjie te kan inhaal?

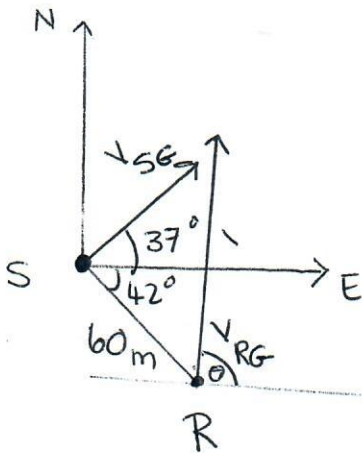
$$\begin{aligned} V_{SGx} &= 13 \text{ km/h} \cos 37^\circ & V_{SGy} &= 13 \text{ km/h} \sin 37^\circ \\ V_{RGx} &= 30 \text{ km/h} \cos \theta & V_{RGy} &= 30 \text{ km/h} \sin \theta \end{aligned}$$

$$V_{SG} = V_{SR} + V_{RG} \quad (\text{See vectors below})$$

$$\therefore V_{SR} = V_{SG} - V_{RG} \quad \text{Therefore } V_{SR} \text{ is directed } 42^\circ \text{ S of E}$$

$$\frac{V_{SRy}}{V_{SRx}} = \tan(-42^\circ) = \frac{V_{SGy} - V_{RGy}}{V_{SGx} - V_{RGx}} = \frac{13 \text{ km/h} \sin 37^\circ - 30 \sin \theta}{13 \text{ km/h} \cos 37^\circ - 30 \cos \theta}$$

$$\text{Solving for } \theta: \theta = 112.9^\circ \Rightarrow 23^\circ \text{ W of N.}$$



The End! / Die Einde!

V_{SG} - velocity of small boat wrt ground
 V_{RG} " " Rescue " " "
 V_{SR} " " small " " Rescue boat

